Review of statistics

Expectation, variance, and standard deviation

In probability theory, if a random variable X can take on n possible values, $X_1, X_2, \ldots, X_n$, with probabilities $\pi_1, \pi_2, \ldots, \pi_n$, then the expected value of X is

$$E(X) = \pi_1 X_1 + \pi_2 X_2 + \ldots + \pi_n X_n,$$

the variance of X is

$$\begin{aligned} \sigma^2(X) &= \pi_1[X_1 - E(X)]^2 \\ &\quad + \pi_2[X_2 - E(X)]^2 + \ldots + \pi_n[X_n - E(X)]^2, \end{aligned}$$

and the standard deviation of X is $\sigma(X) = [\sigma^2(X)]^{1/2}$.

Covariance

In probability theory, the covariance between two random variables X_1 and X_2 is defined as

$$\sigma(X_1, X_2) = E\{[X_1 - E(X_1)][X_2 - E(X_2)]\}$$

and the correlation between X_1 and X_2 is defined as

$$\rho(X_1, X_2) = \frac{\sigma(X_1, X_2)}{\sigma(X_1)\sigma(X_2)}$$

Covariance (Cont.)

The covariance

$$\sigma(X_1, X_2) = E\{[X_1 - E(X_1)][X_2 - E(X_2)]\}$$

is positive if

$$X_1 - E(X_1) \text{ and } X_2 - E(X_2)$$

tend to have the same sign, negative

$$X_1 - E(X_1) \text{ and } X_2 - E(X_2)$$

tend to have opposite signs, and zero if

$$X_1 - E(X_1) \text{ and } X_2 - E(X_2)$$

show no tendency to have the same or opposite signs.

Covariance (Interpretation)

Mathematically, therefore, the covariance

$$\sigma(X_1, X_2) = E\{[X_1 - E(X_1)][X_2 - E(X_2)]\}$$

measures the extent to which the two random variables tend to move together.

Economically, buying two assets with returns that are imperfectly, and especially, negatively correlated is like buying insurance: one return will be high when the other is low and vice versa, reducing the overall risk of the portfolio.

Correlation

The correlation

$$\rho(X_1, X_2) = \frac{\sigma(X_1, X_2)}{\sigma(X_1)\sigma(X_2)}$$

has the same sign as the covariance, and is therefore also a measure of co-movement.

But "scaling" the covariance by the two standard deviations makes the correlation range between -1 and 1:

$$-1 \leq \rho(X_1, X_2) \leq 1$$

A formal proof of $|\rho| \leq 1$ involves Schwartz Inequality

Properties of covariance

- $\mathrm{Cov}(X_1,X_1) = \mathrm{Var}(X_1)$
- $\mathrm{Cov}(a\,X_1,X_2) = a\,\mathrm{Cov}(X_1,X_2)$
- $\mathrm{Cov}(X_1+c,X_2) = \mathrm{Cov}(X_1,X_2)$
- $\mathrm{Cov}(X_1+X_2,X_3) = \mathrm{Cov}(X_1,X_3)+\mathrm{Cov}(X_2,X_3)$

- If $X_3 = X_1+X_2$ then $\mathrm{Var}(X_3) = \mathrm{Cov}(X_3, X_3)$

$$= \mathrm{Cov}(X_1+X_2, X_1+X_2)$$
$$= \mathrm{Cov}(X_1,X_1) + \mathrm{Cov}(X_1,X_2) + \mathrm{Cov}(X_2,X_1) + \mathrm{Cov}(X_2,X_2)$$
$$= \mathrm{Var}(X_1) + \mathrm{Var}(X_2) + 2\mathrm{Cov}(X_1,X_2)$$

- $\mathrm{Var}(aX_1+bX_2)= a^2\,\mathrm{Var}(X_1) + b^2\,\mathrm{Var}(X_2) + 2ab\,\mathrm{Cov}(X_1,X_2)$

Additional properties

Note: notations are changed to X, Y for the ease of writing

$E(X) = \mu_X$ and $E(Y) = \mu_Y$, the covariance between X and Y is defined as

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

This definition implies

$$\begin{aligned} Cov(X, Y) &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E(XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y) \\ &= E(XY) - \mu_X E(Y) - \mu_Y E(X) + \mu_X \mu_Y \\ &= E(XY) - \mu_X \mu_Y - \mu_Y \mu_X + \mu_X \mu_Y \\ &= E(XY) - E(X)E(Y) \end{aligned}$$

Linear regression

For two random variables X and Y. Can always write a linear function:

$$Y=\alpha + \beta X + \varepsilon$$

$$E(\varepsilon)=0$$

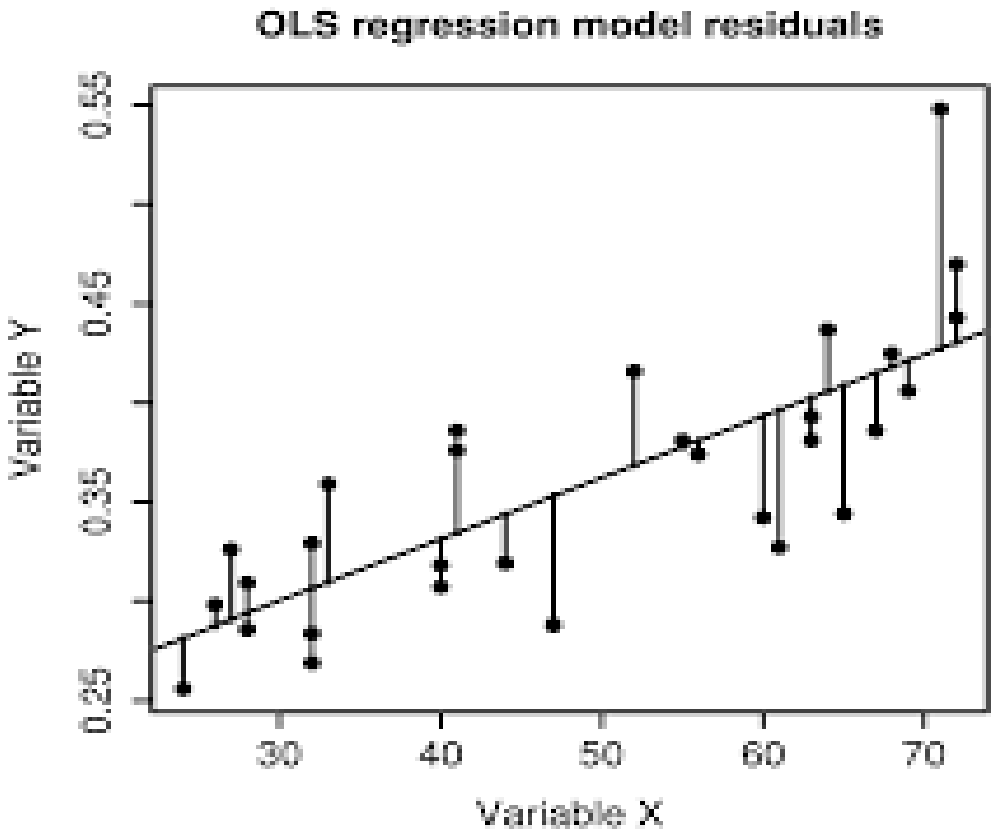
$$Cov(X, \varepsilon) = 0$$

Then $Cov(Y, X) = Cov(\alpha + \beta X + \varepsilon, X)$

$$=0 + \beta\, Cov(X,X) + 0$$

$$= \beta\, Var(X)$$

$$\Rightarrow \beta = Cov(Y, X)/Var(X)$$

Property of linear (or OLS) regression

- β is the number that minimize the Var(ε) if we write Y as a linear function of X
- Graphically: β is the slope of the straight line that minimize $\sum(Y_i - \alpha - \beta * X_i)^2$